

Setup. A deployed teacher policy π_T repeatedly encounters near-identical situations. We store its successful rollouts in a library $\mathcal{L} = \{(k_i, a_i)\}$, where each key $k_i = \varphi(o_i)$ is an intermediate representation extracted from the teacher’s forward pass and each payload a_i is the executed action chunk. At each decision step a scheduler σ selects either REPLAY (execute the nearest neighbor’s payload) or INFER (run the full forward pass), inducing a mixed policy π_σ . The episode outcome is $Y \in \{0, 1\}$ and the teacher-only success rate is SR_T . The paper formalizes the question of how many steps can be replayed without degrading success.

Concept. A step of inference is redundant when the action it produces is already determined by stored experience, so that the forward pass contributes no information beyond what the library holds. This is analogous to inter-frame compression in video coding: a MISS step corresponds to an I-frame, computed in full, and a HIT step to a P-frame, reconstructed from stored data.

Step-level redundancy. Let d_{π_T} denote the state distribution induced by the teacher acting alone, and let $s_t \sim d_{\pi_T}$. Step t is δ -redundant if

$$\left| P(Y=1 \mid \text{do}(a_t = \hat{a}_t^{\text{replay}})) - P(Y=1 \mid \text{do}(a_t = a_t^{\pi_T})) \right| \leq \delta. \quad (1)$$

Since Y is binary, the left-hand side equals the total variation distance between the two outcome distributions. The definition is stated in terms of the outcome rather than action similarity, because similar actions may lead to failure and dissimilar actions may both succeed. The $\text{do}(\cdot)$ operator denotes intervention; conditioning on the observed action would introduce confounding, because the teacher’s choice of action is itself informative about the state. Eq. 1 defines redundancy; the operational quantity is Eq. 3.

Remark (failure of composition). Eq. 1 is evaluated under d_{π_T} . Under deployment, earlier steps may already have been replayed, so the state distribution at step t differs from d_{π_T} and replacement errors accumulate. Per-step redundancy therefore does not compose: individual replaceability of each step does not imply joint replaceability. The operational quantity must be defined at the level of the scheduler and estimated from closed-loop rollouts. Offline metrics computed on a fixed set of teacher trajectories cannot estimate it.

Recoverable redundancy. The replay share of a scheduler is

$$h(\sigma) = \mathbb{E}_{\pi_\sigma} \left[\frac{1}{T} \sum_{t=1}^T \mathbb{1}\{\sigma_t = \text{replay}\} \right], \quad (2)$$

where the expectation is taken under the trajectory distribution induced by π_σ itself, so that closed-loop effects are part of the definition. For a class of schedulers $\mathcal{C}$, define

$$R_{\mathcal{C}}(\varepsilon) = \sup_{\sigma \in \mathcal{C}} h(\sigma) \quad \text{s.t.} \quad \text{SR}(\pi_\sigma) \geq \text{SR}_T - \varepsilon, \quad (3)$$

where ε is the admissible loss in success rate; the supremum need not be attained. Since $R_{\mathcal{C}}$ is a supremum, the replay share of any feasible scheduler is a lower bound on it, and reported values cannot overstate the quantity.

Stratification by information set. Let $\mathcal{C}(\mathcal{I})$ denote the class of causal schedulers whose decisions depend only on the information set $\mathcal{I}$, i.e. on present and past quantities but not on future states or the outcome; without this restriction the supremum is attained by schedulers that condition on Y and Eq. 3 is uninformative. Consider $\mathcal{I}_\varphi \subseteq \mathcal{I}_{\text{sim}} \subseteq \mathcal{I}_{\text{all}}$, where $\mathcal{I}_\varphi$ contains the quantities visible to φ -key retrieval, and $\mathcal{I}_{\text{sim}}$ additionally contains the simulator state. The first inclusion holds because, in simulation, the state determines the observation and hence the key. Then

$$h(\sigma_{\text{thr}}) \leq R_{\mathcal{C}(\mathcal{I}_\varphi)}(\varepsilon) \leq R_{\mathcal{C}(\mathcal{I}_{\text{sim}})}(\varepsilon) \leq R_{\mathcal{C}(\mathcal{I}_{\text{all}})}(\varepsilon) = R(\varepsilon). \quad (4)$$

The first inequality holds because the threshold scheduler σ_{thr} is an element of $\mathcal{C}(\mathcal{I}_\varphi)$; the remaining inequalities hold because enlarging the information set enlarges the feasible set. An empirical comparison between a measured oracle arm and a measured threshold arm does not follow from Eq. 4: the two measurements are lower bounds on different terms.

Attribution.

$$R = R(\varepsilon; \mathcal{M}, \pi_T, \mathcal{L}, \varphi) \quad (5)$$

depends jointly on the task $\mathcal{M}$, the teacher π_T , the library $\mathcal{L}$, and the key map φ . Reported values must specify all four arguments; a statement of the form “this task is 60% redundant” is not well defined.

Monotonicity. For any scheduler class closed under ignoring library entries, for example $\mathcal{C}(\mathcal{I}_{\text{all}})$,

$$\mathcal{L} \subseteq \mathcal{L}' \implies R(\varepsilon; \mathcal{L}) \leq R(\varepsilon; \mathcal{L}'). \quad (6)$$

Proof. A scheduler feasible on $\mathcal{L}$ remains feasible on $\mathcal{L}'$ by ignoring the entries of $\mathcal{L}' \setminus \mathcal{L}$, with identical success rate and replay share; the feasible set therefore grows and the supremum does not decrease. $\square$

The argument requires closure under ignoring entries and does not apply to threshold rules: an added entry may attain a high retrieval score while being a poor match, which changes the induced policy and can reduce the success rate. Monotonicity under threshold rules is therefore an empirical question.

Intrinsic limit.

$$R^*(\varepsilon) = \sup_{\mathcal{L}} R(\varepsilon; \mathcal{L}) \quad (7)$$

depends only on the task and the teacher and upper-bounds $R(\varepsilon; \mathcal{L})$ for every library. A step can be recognized as redundant only if the library contains a sufficiently close experience, so increasing $\mathcal{L}$ increases the recoverable share while R^* is unchanged. R^* is not measurable, since the space of libraries cannot be exhausted; it serves as a reference point and is not estimated.

Relation to calibration. Eq. 3 also describes the system’s calibration procedure. Selecting the threshold on a calibration split to maximize the replay share subject to the success-rate constraint solves Eq. 3 empirically within $\mathcal{C}(\mathcal{I}_\varphi)$; the threshold sweep of experiment E1 therefore traces the empirical Pareto frontier of $R_{\mathcal{C}(\mathcal{I}_\varphi)}(\varepsilon)$. The deployment stages correspond to the same program: collection constructs $\mathcal{L}$, shadow operation estimates the score distributions, calibration solves the program, and serving executes the resulting scheduler.

Experimental program. Each experiment addresses one argument of Eq. 5 or one term of Eq. 4.

- **E1, main frontier** (LIBERO complete; RoboCasa365 in progress). Threshold sweeps measure $h(\sigma_{\text{thr}})$ as a function of ε , the first term of Eq. 4; two benchmarks and two teachers vary the $\mathcal{M}$ and π_T arguments of Eq. 5. Current results: 50% of decisions replayable at <1 pp loss in success rate and 63% at <5 pp on LIBERO-Spatial; 30% at ~ 1 pp on LIBERO-10. The difference between the suites indicates that recoverable redundancy depends on task difficulty, consistent with Eq. 5.
- **E2, oracle arm** (to run). The same system with retrieval by simulator state estimates the second term of Eq. 4. It separates coverage (the library lacks a close experience, so no key can recover the step) from key loss (the experience is present but the index misses it). If the oracle and threshold arms are close, the internal keys lose little information; a large gap quantifies the margin attributable to the index; a low oracle value indicates that the remaining gap is due to coverage.
- **E3, library-size scaling** (complete). R increases with $\mathcal{L}$, from 0.27 to 0.52 on LIBERO-10 as the library grows from 1 to approximately 39 trajectories per task. This is the empirical counterpart of Eq. 6, which does not imply it for threshold rules; the observed monotone increase is a finding about robustness to library growth. Two boundaries: pure replay remains below the teacher at the full data budget, so replay alone is not a viable policy; and no plateau is resolved, so saturation is not claimed.
- **E4, temporal structure** (complete). Hits are not independent across steps. With the stickiness coefficient $s = P(\text{hit}_t \mid \text{hit}_{t-1}) - P(\text{hit})$, which is zero under independence, we measure $P(\text{hit} \mid \text{hit}) = 0.91\text{--}0.96$ against marginal hit rates of $0.28\text{--}0.70$ across seven configurations; median replay runs last 9–12 steps (P90 up to 52); and 93–98% of consecutive hits within a run retrieve the same library trajectory. The last property explains why long replay runs are compatible with the failure of composition: the replayed segment follows a trajectory that previously succeeded, which bounds the deviation introduced by replacement.
- **E5, score discriminability** (offline; to run). The AUROC of the retrieval score with respect to episode outcome measures whether the quantity the threshold operates on is informative. The result is descriptive only: episode-level labels paired with step-level scores establish association, not a counterfactual.
- **E6, cross-scene transfer** (in progress). Varies $\mathcal{M}$ with $(\pi_T, \mathcal{L}, \varphi)$ held fixed: the library is built in one kitchen scene and queried in a held-out scene. Preliminary observations show robustness on large-motion furniture tasks and weakness on precise dynamic tasks; the controlled experiment has not yet been run.
- **E7, key ablations** (offline replay of logged data). Varies φ : internal representations against external encoders, extraction depth (vision tower against mid-backbone), and multi-modal fusion (per-field weighting against a single concatenated key). This addresses whether the teacher’s own representation is an adequate index.
- **E8, warm start** (stale data; to re-run). Extends the scheduler’s action space from the binary choice in Eq. 2 to a third option: a partially redundant step initializes the flow integration with the retrieved action and skips the early denoising steps. The control compares the full teacher, blind step reduction, and warm start at an equal number of skipped steps.
- **E10, trained-router comparison** (in progress). A learned router is an element of $\mathcal{C}(\mathcal{I}_\varphi)$, so its replay share is a further lower bound on the same term of Eq. 4. The comparison asks whether the threshold rule is already close to the reachable set of its class; if the learned scheduler does not raise the bound appreciably, the threshold is empirically near-optimal within $\mathcal{C}(\mathcal{I}_\varphi)$.

All reported values are lower bounds by construction (Eq. 3); the gaps in Eq. 4 quantify the margin available to better schedulers.